Productive Out Runs Above Average (PORAA): Valuing the Outs Hitters Make, 1908–2025

Research paper and dataset description

Abstract

Most batting metrics reward what happens when a hitter reaches base and treat the outs he makes as interchangeable. This paper introduces Productive Out Runs Above Average (PORAA), a run-value statistic that scores every batter-made out by the change in run expectancy it produces, relative to the average out in the same season and base-out state. Using Retrosheet play-by-play data for 199,897 American and National League regular-season games from 1908 through 2025 (9.66 million batter outs, 77,939 batter-seasons), we find that PORAA is real but modest. The standard deviation across batter-seasons with at least 400 plate appearances is 2.65 runs, with extremes near ±10 to 12 runs. Double-play avoidance is the dominant component (correlation with PORAA of −0.59), strikeout rate matters little (−0.14), and strikeout and double-play rates together explain about 40 percent of the per-out variation. A residual component persists from year to year (r = 0.21). Single-season reliability is low (split-half r = 0.17 for 300+ outs, about 0.29 adjusted to a full season), so season totals should be read cautiously.

1. Introduction

Run expectancy matrices (the RE24 framework) assign a value to each of the 24 base-out states: the average number of runs scored from that state to the end of the inning. Hits and walks change the state favorably and are credited by metrics such as wOBA and WAR. Outs are generally charged a roughly uniform penalty. Yet outs differ. A sacrifice fly scores a run. A groundout can advance a runner. A double play removes a baserunner and an out together. A strikeout does neither.

The tempting shortcut is to credit any out that advances a runner. That is not correct. Moving a runner from second to third with one out can leave a team with fewer expected runs than leaving the runner on second with no outs. An out can therefore be “productive” in the traditional sense and still lower run expectancy. PORAA avoids this error by measuring the actual change in run expectancy and judging it against what the average out in that same situation produces, never against doing nothing.

The question PORAA answers is narrow and practical: given that a hitter is going to make an out in this situation, how good is the out he makes?

2. Data

The data are Retrosheet event files as distributed by the Chadwick Baseball Bureau, parsed with Chadwick’s `cwevent` tool, which produces one record per play with base-runner identities, event codes, outs on the play, and the destination of the batter and each runner.

- American and National League regular-season games, 1908–2025. All-star games, postseason games, and Negro League and other non-AL/NL files are excluded.
- 199,897 games, parsed from team event files filtered through each season's team table.
- Sanity checks: the 2023 run total reproduces the published 22,432 runs; run expectancy with the bases empty and no outs is about 0.38 in 1908, 0.52 in 2023, and 0.50 in 2025.

Player names come from the Retrosheet biographical file and are given names (Edward Collins is Eddie Collins).

3. Method

3.1 Run expectancy

For each season we estimate RE from the average runs scored from each base-out state to the end of the half-inning, using only complete half-innings (three outs recorded) in innings one through eight. This excludes walk-offs, called games, and late-inning situations in which incentives differ.

3.2 Run value of an out

For a batter out, ΔRE = runs scored on the play + RE(end state) − RE(start state), where RE of the three-out state is zero.

3.3 Definition of PORAA

PORAA = Σ [ΔRE(out) − E[ΔRE | all batter outs, season, base-out state]]

The baseline is the average ΔRE across all batter outs in the same season, base state, and number of outs. Three choices are deliberate:

- **All outs count.** Filtering to productive outs would build the answer into the sample and would stop strikeouts from ever counting against anyone.
- **The baseline is the average out, not the average plate appearance.** Comparing against all plate appearances would reproduce RE24, in which hits and walks overwhelm everything.
- **The baseline spans all out types.** An earlier design compared each out only with outs of the same type. That version is mean-zero by construction for strikeouts and double plays and removes the main signal. We report a within-type variant (PORAA_within_type) as a secondary measure of execution given the type of out.

No leverage weighting is applied, so PORAA does not become a clutch measure.

3.4 What counts as a batter out

A batter out is a plate-appearance-ending generic out or strikeout in which the batter is retired: groundouts, flyouts, lineouts, popups, sacrifice flies and bunts, and double and triple plays. It excludes reaching on an error, fielder's choices in which the batter is safe, strikeouts in which the batter reaches on a wild pitch or passed ball, and interference. Baserunning outs that do not result from a batter-made out belong to baserunning metrics. Because the baseline is the same season and state cell, league total PORAA is zero by construction. Cells with fewer than 30 outs are flagged.

4. Results

4.1 Value by type of out

Across 9,658,891 batter outs, average PORAA by type is shown in Figure 1 and the table below.

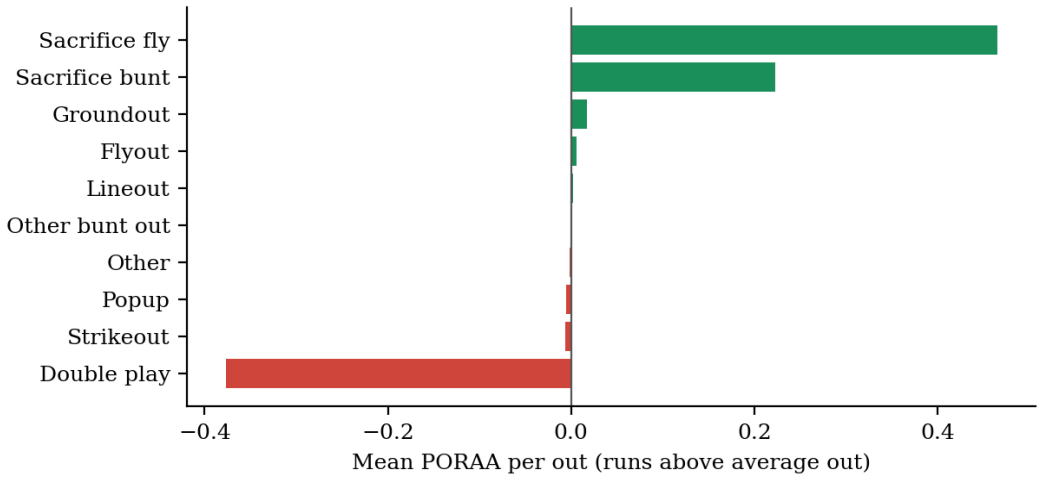


Figure 1. Mean PORAA per out by type of out, all seasons, 1908–2025.

Type of out	Outs	Mean PORAA (runs)	Mean ΔRE (runs)
Sacrifice fly	98,943	+0.465	-0.048
Sacrifice bunt	161,128	+0.223	-0.169
Groundout	2,908,264	+0.017	-0.224
Flyout	2,664,749	+0.006	-0.261
Lineout	478,036	+0.002	-0.263
Other bunt out	39,772	-0.000	-0.265
Other	222,005	-0.002	-0.224
Popup	520,637	-0.005	-0.289
Strikeout	2,216,667	-0.006	-0.280
Double play	348,690	-0.377	-0.819

Three points stand out. First, the double play is by far the costliest event: each is worth about 0.38 runs below the average out in its state. Second, strikeouts are close to average, because in most states the average out is itself frequently a strikeout. Third, successful sacrifice bunts grade positive against the average out (+0.22), contrary to a common expectation that bunting always costs runs. A sacrifice bunt does lower run expectancy in absolute terms (average ΔRE of −0.17), but the typical out in the same situation lowers it by more. This is a success-only comparison: failed attempts are scored as strikeouts, popups, or are not outs at all, so it is not an argument that bunting beats swinging.

With the bases empty, outs differ almost not at all in run value, so empty-base outs contribute little. With two outs, any out ends the inning and ΔRE is identical for every out

type.

4.2 Spread across hitters

For batter-seasons with at least 400 plate appearances (17,089 seasons), the standard deviation of PORAA is 2.65 runs, and the 1st and 99th percentiles are −6.5 and +6.1 runs (Figure 2). By decade the standard deviation ranges from 2.37 (2020s) to 3.04 (1920s). The extremes are larger than we first expected (we guessed ±3 to 5 runs).

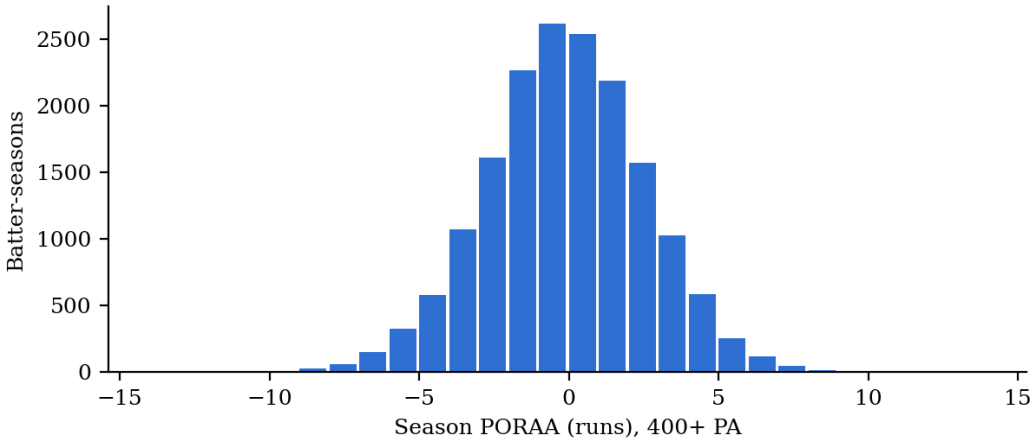


Figure 2. Distribution of season PORAA among batter-seasons with 400+ PA.

Best single seasons (400+ PA): Tom Herr 1985 (+12.3), Bob Wilfong 1979 (+12.2), Charlie Gehringer 1935 (+11.2), Stan Harris 1924 (+10.0). Worst: David Ortiz 2011 (−10.7), Carlos Correa 2023 (−10.3), Harmon Killebrew 1970 (−10.2). Career leaders (3,000+ PA): Eddie Collins (+69.5), Fred Walker (+53.9), Sam West (+50.0). Lowest: Ernie Lombardi (−79.5), Ivan Rodriguez (−59.3). Lombardi's figure is consistent with his well-known double-play totals. Appendix A lists the top and bottom ten.

5. Validation

5.1 Reliability

Splitting each batter-season into alternating outs and correlating the two halves gives a split-half reliability of 0.17 (12,696 batter-seasons with at least 300 scored outs). The Spearman–Brown adjustment to a full season gives about 0.29. The observed year-to-year correlation of PORAA per 100 outs, for hitters with 400+ PA in consecutive seasons (11,806 pairs), is 0.28, in line with the adjusted split-half figure. The year-to-year correlation of the within-type variant is lower (0.12).

5.2 Relationship to strikeouts and double plays

Among hitters with 400+ PA, PORAA correlates −0.59 with the share of outs that are double plays and −0.14 with the share that are strikeouts (Figure 3). A regression of PORAA per 100 outs on those two rates alone explains 40 percent of the variance. Most of the measurable signal is double-play avoidance.

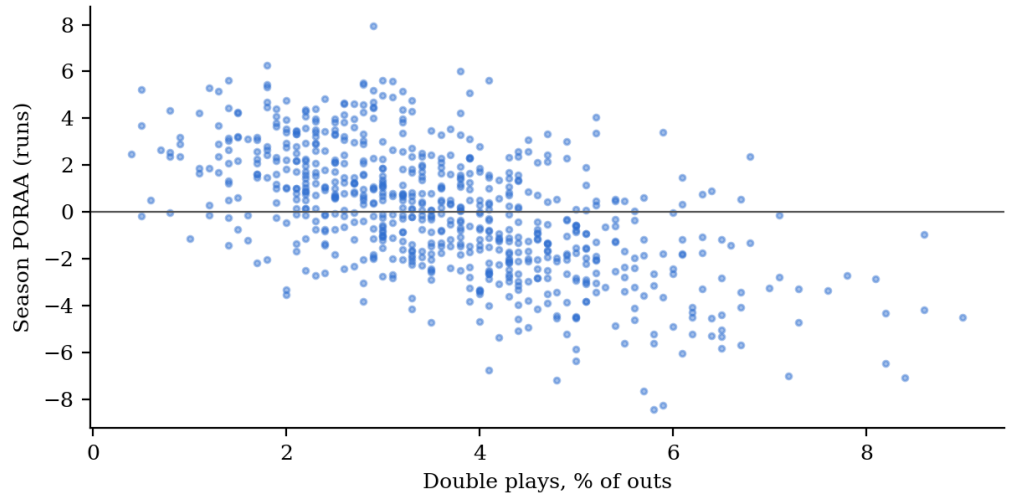


Figure 3. Season PORAA against double-play share of outs (random sample of 700 batter-seasons, 400+ PA).

The importance of double plays is stable across eras even though the run environment is not (Figure 4): strikeouts rose from about 13 percent of batter outs in the 1900s to about 33 percent in the 2020s, while double plays stayed near 3 to 4 percent.

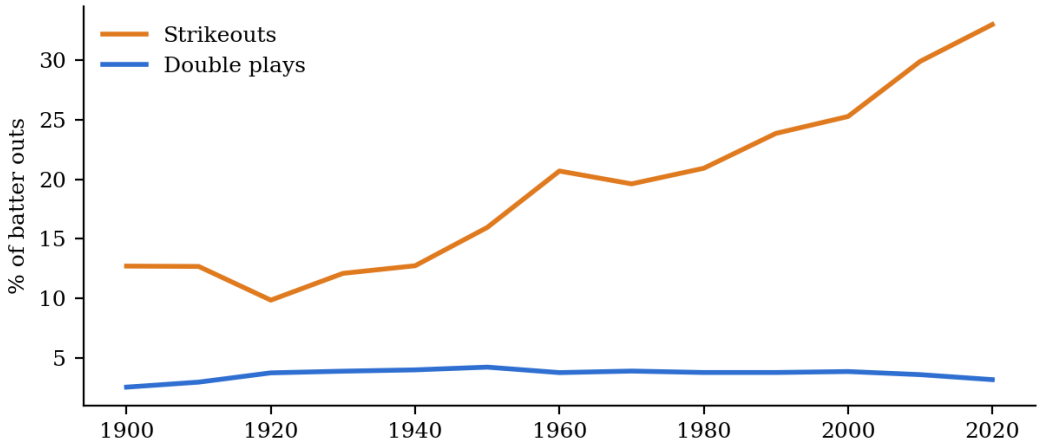


Figure 4. Strikeouts and double plays as a share of batter outs, by decade.

5.3 Is there anything beyond K and GIDP?

After removing the strikeout and double-play components, the residual still correlates 0.21 from one season to the next. That is a small but non-zero persistent component, consistent with a real skill such as directing contact with runners on or reading situations. It could also reflect runner speed, ballpark effects, or lineup context, none of which is adjusted for here.

6. Limitations

- **No runner or batted-ball adjustment.** Whether a runner advances on a groundout depends partly on the runner’s speed and reads, and results depend on where the ball lands.

- **Success-only samples** for bunts and fielder's choices: batter-safe plays are not outs and are not scored.
- **Era and rule changes.** The extra-inning automatic runner (2020 onward) is scored against the regular RE matrix; early seasons have smaller cell samples; shortened seasons call for per-100-outs rates.
- **Same-sample baselines.** League totals are zero by construction; cross-fitting would be better for formal tests.
- **Noise.** With full-season reliability near 0.3, individual seasons are weak evidence about a hitter.
- **Names** are Retrosheet given names and may differ from familiar names.

7. Future work

- Adjust the baseline for runner sprint speed and the batter's batted-ball profile (Statcast era).
- Regress observed PORAA toward a hitter-specific expectation for stabilized estimates.
- Test whether pull-side contact with a runner on second explains the persistent residual.
- Decompose PORAA by base-out state to find where persistent skill lives.
- Pool adjacent seasons to stabilize small cells in the early years.

Appendix A. Leaders

Single seasons (400+ PA), best and worst

Best	Season	PORAA
Thomas Herr	1985	+12.3
Robert Wilfong	1979	+12.2
Charles Gehringer	1935	+11.2
Stanley Harris	1924	+10.0
Thomas Herr	1987	+9.7
Carson Bigbee	1922	+9.7
Timothy Foli	1982	+9.4
Frank Frisch	1921	+9.3
Joseph Vaughan	1937	+8.7
Clint Barmes	2006	+8.7

Worst	Season	PORAA
David Ortiz	2011	-10.7
Carlos Correa	2023	-10.3
Harmon Killebrew	1970	-10.2
Antonio Armas	1983	-10.0
Dale Murphy	1988	-9.8
Ronald Santo	1973	-9.6
Charles Knight	1987	-9.5
Arthur Darwin	1972	-9.4
John Lollar	1959	-9.4
Joseph Cronin	1935	-9.3

Careers (3,000+ PA), best and worst

Best	Last season	PORAA
Edward Collins	1930	+69.5
Fred Walker	1949	+53.9
Samuel West	1942	+50.0
Charles Gehringer	1942	+48.4
William Surhoff	2005	+48.2
Johnny Damon	2012	+45.8
Henry Manush	1939	+45.1
Robert Veach	1925	+43.5
Tristram Speaker	1928	+42.3
Joseph Judge	1934	+41.2

Worst	Last season	PORAA
Ernesto Lombardi	1947	-79.5
Ivan Rodriguez	2011	-59.3
Joseph Torre	1977	-56.7
James Rice	1989	-53.4
Paul Konerko	2014	-47.5
William Horton	1980	-46.9
Javier Lopez	2006	-46.2
Harmon Killebrew	1975	-46.0
Lee May	1982	-45.8
Lance Parrish	1995	-45.6

Appendix B. Dataset files

File	Contents
PORAA_player_season.csv	One row per batter-season: PORAA, within-type PORAA, runners-on / RISP / runner-on-2B splits, K, GIDP, PA, per-100-outs and per-600-PA rates
PORAA_career.csv	Career totals
PORAA_play_level.csv.gz	Every scored batter out (9.66M rows): base-out state, RE before/after, ΔRE, baseline, PORAA
PORAA_run_expectancy_matrix.csv	Season-specific RE24 matrix with cell counts
PORAA_baselines.csv	Average ΔRE of outs by season, base state, and outs
PORAA_coverage_by_season.csv	Games, PA, outs, and sanity checks by season
poraa_build.py	Build script

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The information used here was obtained free of charge from and is copyrighted by Retrosheet. Interested parties may contact Retrosheet at www.retrosheet.org. Event files were obtained from the Chadwick Baseball Bureau’s Retrosheet mirror and parsed with Chadwick’s cwevent.